

A Motion-Aware Dynamic Attention for Efficient Anomaly Detection

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Abstract

Efficient and accurate video anomaly detection is critical for real-time applications such as autonomous driving, large-scale surveillance, and security monitoring. However, many existing approaches rely on implicit motion representations, which often struggle to capture complex, subtle, or rapidly evolving anomalies, limiting both detection accuracy and computational efficiency. To address this gap, we propose *MATE* (*A Motion-Aware Attention and Temporal-Conditioned Convolution Encoder*), a lightweight motion-conditioned architecture that explicitly integrates motion magnitude and directional information throughout the network. *MATE* leverages directional motion maps, motion-gated multi-scale temporal-conditioned convolutions, motion-modulated kernelized linear attention, and motion-aware temporal pooling to focus computation on anomaly-relevant regions while remaining highly parameter-efficient. On the UCF-Crime benchmark under the standard video-level evaluation protocol, *MATE*-FAST and *MATE*-BASE achieve ROC-AUC scores of 89.75% and 91.78%, respectively, outperform recent strong baselines with substantially fewer parameters (FAST 8.5k, BASE 40k). Extensive ablation studies, synthetic signal-to-noise ratio experiments, and per-fold cross-validation demonstrate that explicit motion conditioning improves temporal feature discriminability and embedding geometry under weak supervision. These results position *MATE* as a practical and effective solution for low-resource and low-latency video anomaly detection. Code and pretrained models are publicly available at <https://github.com/AIx699/MATE>

1. Introduction

Detecting anomalies in videos is extremely critical for safety-sensitive and time-constrained scenarios, such as autonomous driving and large-scale public real-world surveillance applications. Real-world anomalies often manifest as subtle or complex spatiotemporal patterns, making detection

challenging, particularly under weak supervision where only video-level labels are available. Recent methods leverage efficient spatiotemporal attention and motion-aware token selection but rely on scalar motion cues, static temporal conditioning, and post-hoc motion integration. These limitations reduce adaptability to rapidly evolving anomalies and hinder discriminative temporal feature learning

Several prior works have addressed these challenges. Robust Temporal Feature Magnitude learning (RTFM) formulates the problem in a multiple instance learning framework and learns robust feature magnitudes that distinguish rare abnormal snippets from dominant normal ones (Jiang et al., 2021). Weakly Supervised Anomaly Localization (WSAL) introduces clustering and normalcy suppression mechanisms to better separate feature clusters for normal and anomalous regions (Zaheer et al., 2020). Self-Supervised Sparse Representation (S3R) decomposes features into sparse representations to isolate normal behavior (Öztürk & Can, 2021). The Magnitude-Contrastive Glance-and-Focus Network (MGFN) enhances spatial-temporal integration via contrastive learning (Liu et al., 2022), while BN-WVAD leverages batch normalization statistics for robust snippet-level anomaly recognition (Zhou et al., 2023; Zhang et al., 2023). More recent work, such as STEAD, focuses on efficient spatiotemporal modeling to balance detection performance and computational cost (Gao & Liu, 2025).

To address the limitations of these approaches, we introduce *MATE*, a motion-aware dynamic attention model that explicitly incorporates motion magnitude and direction throughout the network. Directional motion maps capture temporal deviations accurately, multi-scale temporal conditioned convolutions adapt features based on local and global context, motion-modulated linear attention efficiently models long-range dependencies, and motion-aware temporal pooling aggregates the most relevant frames into a compact video representation. By integrating these components, *MATE* maintains near-linear computational complexity while learning motion-sensitive, discriminative features.

We evaluate *MATE* on the UCF-Crime dataset (Sultani et al., 2018), achieving an ROC-AUC of 91.78% with only 40k parameters, outperforming previous state-of-the-art methods. Motion map visualizations and embedding analyses confirm that explicit motion integration improves interpretability

and geometric separation of normal and anomalous events, demonstrating the model’s ability to focus on anomaly-relevant spatiotemporal regions under weak supervision.

2. Related Work

Video Feature Extractors

Learning effective video representations is fundamental for video understanding. Early approaches used 3D convolutional networks (3D CNNs) to directly model spatiotemporal features (Carreira & Zisserman, 2017). While effective, 3D CNNs are computationally intensive, motivating efficient variants such as X3D (Feichtenhofer, 2020), which progressively expand network dimensions (depth, width, resolution) to balance accuracy and efficiency. Similarly, SlowFast networks (Feichtenhofer et al., 2019) adopt dual pathways to separately capture slow semantic and fast motion features. In our approach, we adopt frozen X3D features for spatial-temporal encoding, and complement them with UTS (Universal Temporal Sampling) features to obtain high-quality, precomputed spatiotemporal representations. By operating on frozen backbones, our method reduces end-to-end training costs while retaining flexibility for motion-sensitive modeling.

Motion Modeling in Video Understanding

Explicit motion modeling is critical for capturing temporal dynamics. Two-Stream Networks (Simonyan & Zisserman, 2014) use optical flow to represent motion, while Temporal Segment Networks (TSN) (Wang et al., 2016) leverage sparse frame sampling to capture long-range temporal dependencies. More recently, TDNet (Wang & et al., 2021) and TEA (Li & et al., 2020) demonstrate that lightweight temporal differencing can approximate motion information without heavy optical flow computation. Unlike these methods, our approach computes motion saliency through differentiable frame differencing (MotionMap), which directly guides attention and convolutional operations, achieving motion-aware modeling without explicit flow computation.

Spatiotemporal Attention for Video

Transformers have achieved strong performance in video modeling by capturing long-range dependencies. Models such as TimeSformer (Bertasius et al., 2021), ViViT (Arnab et al., 2021), and Video Swin (Liu & et al., 2022) use separable temporal and spatial attention to reduce computation and maintain accuracy. Despite success, quadratic attention remains a bottleneck, particularly for long video sequences. Our Motion-Modulated Efficient Dynamic Attention (MM-EDA) addresses this by employing linear-time kernelized attention, conditioned on motion saliency, enabling scalable temporal and spatial dependency modeling.

Efficient and Linear Attention Mechanisms

Linear attention techniques approximate standard attention with linear complexity in sequence length, facilitating scalability to longer sequences. Performer (Choromanski et al., 2021) uses positive kernel feature maps to achieve linear-time attention, while Linear Transformers (Katharopoulos et al., 2020b) and Nyströmformer (Xiong et al., 2021) leverage low-rank or kernel-based approximations. Building upon these ideas, MM-EDA modulates query representations with motion weights, ensuring that motion-salient regions receive higher attention while maintaining linear complexity in both temporal and spatial dimensions.

Motion-Aware Pooling and Token Aggregation

Motion-guided temporal pooling has emerged as an effective technique for compressing video representations. Query-based methods such as Set Transformer (Lee et al., 2019) and TokenLearner (Ryoo et al., 2021) dynamically select tokens for efficient aggregation. Our Motion-Aware Temporal Pooling extends this idea by weighting frame descriptors according to the differentiable MotionMap, followed by learnable query attention, producing a compact, motion-sensitive video embedding suitable for downstream prediction tasks.

Positioning of Our Approach

Overall, prior work highlights the importance of effective backbones, motion modeling, and efficient attention mechanisms. Our approach differs in three key aspects: (i) we leverage UTS and pretrained X3D features for scalable feature extraction, (ii) we introduce differentiable MotionMap-guided convolution, attention, and temporal query pooling, and (iii) we employ linear-time spatiotemporal attention for motion-adaptive, efficient video modeling. By integrating these components, our method balances accuracy, efficiency, and motion sensitivity, making it suitable for real-time and resource-constrained applications.

3. Methodology

3.1. X3D Feature Extraction and Video Preprocessing

3.1.1. VIDEO INPUT AND CLIP SAMPLING

Each video $\mathbf{V}$ is represented as a sequence of RGB frames:

$$\mathbf{V} = \{\mathbf{I}_t\}_{t=1}^{T_{\text{full}}}, \quad \mathbf{I}_t \in \mathbb{R}^{3 \times H_0 \times W_0},$$

where T_{full} is the total number of frames, and H_0, W_0 are the original spatial dimensions. Videos are divided into fixed-length clips of 16 frames using uniform sampling. Each clip $\mathbf{C}^{(k)}$ is:

$$\mathbf{C}^{(k)} = \{\mathbf{I}_{t_1^{(k)}}, \dots, \mathbf{I}_{t_{16}^{(k)}}\}.$$

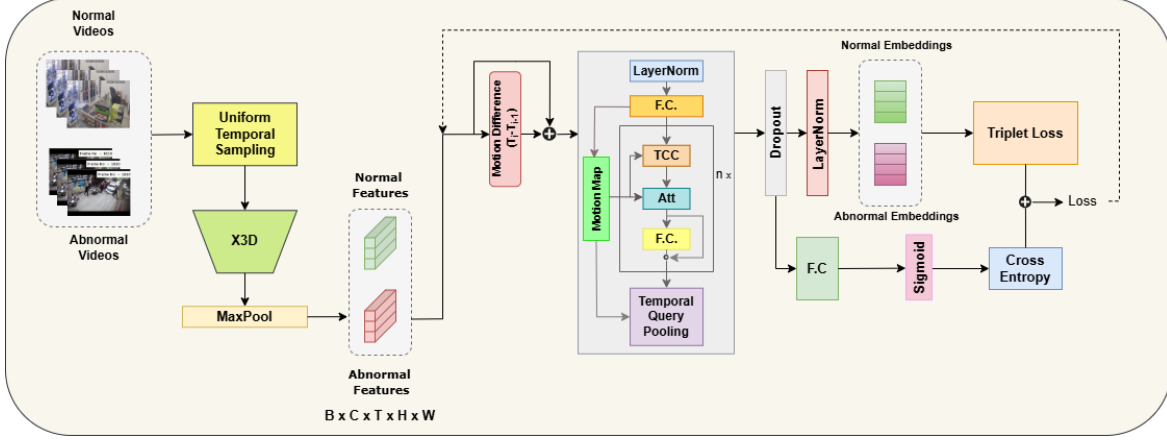


Figure 1. Overview of the proposed motion-aware video representation learning architecture. Input video clips are temporally sampled and processed by X3D to extract features. Motion differences are fused with backbone embeddings, optimized jointly using triplet and classification losses.

3.1.2. PREPROCESSING TRANSFORMATIONS

Each input video clip undergoes standard preprocessing. Frames are first temporally subsampled, and pixel intensities are normalized to the range $[0, 1]$. Channel-wise normalization is then applied with $\mu = (0.45, 0.45, 0.45)$ and $\sigma = (0.225, 0.225, 0.225)$. Finally, the clip is resized such that the short side matches the target resolution and center-cropped to 320×320 . After preprocessing, each clip tensor has shape

$$\mathbf{X}^{(k)} \in \mathbb{R}^{3 \times 16 \times 320 \times 320}.$$

3.1.3. X3D FEATURE EXTRACTION AND STORAGE

A pretrained X3D model is truncated before the classification head. Each clip is forwarded through the backbone to produce a spatiotemporal feature tensor:

$$\mathbf{Z}^{(k)} \in \mathbb{R}^{1 \times 192 \times 16 \times 10 \times 10},$$

where 192 is the channel dimension, 16 is the temporal resolution, and 10×10 is the spatial resolution. Clip-level features are saved independently as NumPy arrays, preserving temporal and spatial structure for downstream motion-aware processing.

3.2. Problem Setup

We consider supervised video-level prediction from video clips. Given an input video tensor

$$\mathbf{X}_0 \in \mathbb{R}^{B \times C_{in} \times T \times H \times W},$$

our objective is to learn a function that maps each video to a scalar prediction, while efficiently modeling spatiotemporal dependencies and emphasizing motion-salient regions.

$$f_{\theta} : \mathbb{R}^{C_{in} \times T \times H \times W} \rightarrow [0, 1], \quad (1)$$

3.3. Motion-Augmented Input Representation

To explicitly encode short-term temporal dynamics, we augment the raw input with a first-order temporal difference:

$$\Delta \mathbf{X}_t = \begin{cases} 0, & t = 0 \\ |\mathbf{X}_t - \mathbf{X}_{t-1}|, & t > 0 \end{cases} \quad (2)$$

The augmented representation is formed by channel-wise concatenation:

$$\mathbf{X}_1 = \text{Concat}(\mathbf{X}_0, \Delta \mathbf{X}) \in \mathbb{R}^{B \times 2C_{in} \times T \times H \times W}.$$

This operation injects first-order temporal differences to efficiently model motion, highlighting motion-salient regions without the overhead of optical flow.

3.4. Feature Normalization and Projection

We permute the tensor to a channel-last format and apply layer normalization followed by linear projection:

$$\mathbf{X} = \text{LN}(\mathbf{X}_1) \mathbf{W}_{in}, \quad \mathbf{W}_{in} \in \mathbb{R}^{2C_{in} \times C}.$$

The resulting feature tensor serves as the shared embedding space for all subsequent modules.

$$\mathbf{X} \in \mathbb{R}^{B \times T \times H \times W \times C}$$

3.5. MotionMap: Temporal Saliency Estimation

To estimate motion saliency, we compute a local temporal mean over a window of size K and the signed temporal deviation:

$$\bar{\mathbf{X}}_t = \frac{1}{K+1} \sum_{i=0}^K \mathbf{X}_{t-i}, \quad \mathbf{F}_t = \mathbf{X}_t - \bar{\mathbf{X}}_t.$$

The channel-wise magnitude is calculated and normalized globally:

$$M_{t,h,w} = \|\mathbf{F}_{t,h,w}\|_2, \quad \tilde{M} = \frac{M}{\max(M) + \varepsilon}. \quad (3)$$

The resulting MotionMap

$$\tilde{M} \in \mathbb{R}^{B \times T \times H \times W \times 1}$$

provides a differentiable approximation of motion importance without explicit optical flow computation.

3.6. Temporal Conditioned Convolution (TCC)

3.6.1. SPATIAL BOTTLENECK

Each frame is processed by a shared spatial convolution:

$$\mathbf{Y} = \text{Conv}_{k \times k}(\mathbf{X}) \in \mathbb{R}^{B \times T \times H \times W \times C_b}.$$

3.6.2. MULTI-SCALE TEMPORAL CONDITIONING

We first compute frame-wise global descriptors via spatial average pooling:

$$\mathbf{g}_t = \frac{1}{HW} \sum_{h,w} \mathbf{X}_{t,h,w}, \quad \mathbf{g}_t \in \mathbb{R}^{B \times C}.$$

To capture temporal context at multiple resolutions, we aggregate global descriptors over different temporal extents:

$$\mathbf{z} = [\mathbb{E}_t(\mathbf{g}_t), \mathbb{E}_{t:t+1}(\mathbf{g}), \mathbb{E}_{t:t+3}(\mathbf{g})] \in \mathbb{R}^{B \times 3C},$$

where $\mathbb{E}_t(\cdot)$ denotes global temporal averaging, while $\mathbb{E}_{t:t+k}(\cdot)$ represents short-range and mid-range temporal pooling over windows of size $k + 1$.

The concatenated multi-scale temporal descriptor $\mathbf{z}$ is then projected to obtain a channel-wise modulation vector:

$$\mathbf{w} = \sigma(\mathbf{z} \mathbf{W}_{tc}), \quad \mathbf{W}_{tc} \in \mathbb{R}^{3C \times C_b}.$$

This modulation extends squeeze-and-excitation style channel recalibration to the temporal domain.

3.6.3. MOTION-GATED RESIDUAL CONNECTION

We introduce a scalar motion gate:

$$\alpha = \sigma(\mathbf{W}_g \cdot \mathbb{E}[\tilde{M}]), \quad (4)$$

which controls the residual contribution:

$$\mathbf{X}' = \mathbf{X} + \alpha \cdot \text{Conv}_{1 \times 1}(\mathbf{Y}). \quad (5)$$

This mechanism yields motion-adaptive convolution with negligible computational overhead.

3.7. Motion-Modulated Efficient Dynamic Attention (MM-EDA)

3.7.1. TEMPORAL ATTENTION

Spatial features are pooled into temporal tokens:

$$\mathbf{s}_t = \frac{1}{HW} \sum_{h,w} \mathbf{X}'_{t,h,w}.$$

We adopt kernelized linear attention using a positive feature map (Choromanski et al., 2021; Wang et al., 2020):

$$\phi(x) = \text{ELU}(x) + 1,$$

Motion weights are defined as:

$$\omega_t = 1 + \tanh\left(\frac{1}{HW} \sum_{h,w} \tilde{M}_{t,h,w}\right). \quad (6)$$

Queries are modulated as $\tilde{Q}_t = \omega_t \cdot \phi(Q_t)$, yielding:

$$\mathbf{O}_t = \frac{\tilde{Q}_t \sum_{t'} \phi(K_{t'}) V_{t'}}{\tilde{Q}_t \sum_{t'} \phi(K_{t'})}.$$

This follows the linear attention formulation of Performer (Choromanski et al., 2021), which approximates full self-attention with linear complexity.

3.7.2. SPATIAL ATTENTION

Temporal outputs are broadcast to spatial tokens and processed via spatial linear attention, where per-pixel motion magnitudes modulate the queries. The resulting computational complexity is:

$$\mathcal{O}(BTHWd),$$

substantially lower than quadratic self-attention.

3.8. Motion-Aware Temporal Pooling

To obtain a compact video-level representation, we compute motion-weighted frame descriptors:

$$\mathbf{f}_t = \sum_{h,w} \frac{\tilde{M}_{t,h,w}}{\sum_{h,w} \tilde{M}_{t,h,w}} \mathbf{X}_{t,h,w}. \quad (7)$$

Learnable temporal queries attend over frames:

$$\alpha_{q,t} = \text{softmax}(\mathbf{Q}_q^\top \mathbf{f}_t), \quad \mathbf{p}_q = \sum_t \alpha_{q,t} \mathbf{f}_t.$$

(Query-based pooling, inspired by the Set Transformer (Lee et al., 2019))

The final prediction is computed as:

$$\hat{\mathbf{p}} = \text{LN}(\mathbf{p}), \quad \hat{y} = \sigma(\hat{\mathbf{p}}_{\text{cls}}). \quad (8)$$

3.9. Loss Functions

Binary Cross-Entropy (BCE): Encourages correct class predictions for each video sample:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (9)$$

Triplet Loss (Hardest Negative): Encourages anchors to be closer to positives than the hardest negatives:

$$\mathcal{L}_{\text{triplet}} = \frac{1}{N} \sum_{i=1}^N \max \left(0, \|a_i - p_i\|_2 - \min_{n_j \in N_i} \|a_i - n_j\|_2 + m \right) \quad (10)$$

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \mathcal{L}_{\text{triplet}} \quad (11)$$

Here, $\hat{y}_i$ denotes the predicted probability, y_i is the ground-truth label, a_i is the anchor feature, p_i the positive feature, n_j negatives, N_i the set of all negatives for anchor i , m the margin, and α a weighting factor. BCE ensures accurate class prediction, while triplet loss enforces feature separation.

4. Experiments

4.1. Dataset and Labeling Protocol

We evaluate our approach on two widely adopted benchmarks for video anomaly and violence detection: UCF-Crime (Sultani et al., 2018) and XD-Violence (Wu et al., 2020). The UCF-Crime dataset is a large-scale, real-world surveillance corpus with 1,900 long untrimmed videos totaling approximately 128 hours, covering diverse indoor and outdoor scenarios. It includes a wide range of anomalous events such as abuse, arson, assault, road accidents, burglary, explosion, fighting, robbery, shooting, stealing, shoplifting, and vandalism, alongside normal activities. Videos are provided under weak supervision, making representation learning and temporal reasoning challenging. The dataset is split into training and test sets, and performance is typically measured using the area under the ROC curve (AUC) on the curated test set (Sultani et al., 2018; Wu et al., 2020).

XD-Violence is a large-scale multimodal violence detection benchmark comprising 4,754 untrimmed videos with audio, spanning about 217 hours from diverse real-world and multimedia sources. Each video is labeled at the video level as violent or non-violent, allowing weakly supervised training. It covers various types of violence, including physical altercations, car accidents, explosions, shootings, and riots, and is partitioned into training and test subsets. Performance is evaluated using average precision (AP) on the curated test set (Wu et al., 2020). Together, these datasets provide

complementary evaluation scenarios: UCF-Crime emphasizes real-world surveillance anomaly detection, while XD-Violence tests audiovisual violence detection in complex, multi-scene content, enabling assessment of model generality and robustness across weakly supervised video tasks.

4.2. Feature Extraction and Preprocessing

All experiments use spatiotemporal features from I3D (Carreira & Zisserman, 2017) pretrained on Kinetics-400 and X3D (Feichtenhofer, 2020). For X3D, videos are sampled at 30 FPS, segmented into 16-frame clips with temporal stride 5, resized to 320 pixels on the short side, center-cropped to 320×320, and channel-normalized. Each clip is passed through the frozen X3D encoder to produce spatiotemporal features preserving temporal and spatial structure, which are uniformly sampled along the video and stored for downstream training. For I3D, clips are forwarded through the frozen encoder to capture rich motion features, using 10 crops for UCF-Crime and 5 crops for XD-Violence to account for spatial variations.

4.3. Training and Evaluation Setup

For both UCF-Crime and XD-Violence, we adopt a weakly supervised setup where videos are first passed through the respective feature extractor to obtain clip-level features. For X3D, the extractor outputs tensors of shape (C, T, H, W) , which are temporally max-pooled during training to $T = 16$, yielding features of shape $(192, 16, 10, 10)$. For I3D, the encoder produces (N_{frames}, D) with $D = 1024$, and we apply average pooling over D to obtain $(512, N_{\text{frames}}, 1, 1)$.

During evaluation, pretrained checkpoints are loaded to compute scores. Training uses balanced batches of size 16 (8 normal, 8 abnormal) and is optimized with AdamW (learning rate 0.002, weight decay 0.002) using a cosine schedule with linear warm-up over 1 epoch. Dropout of 0.4 for BASE and 0.29 for FAST is applied throughout. For UCF-Crime, stratified 5-fold cross-validation is performed to assess stability across splits. All experiments are implemented in PyTorch on a single GPU with fixed random seeds for reproducibility.

4.4. Comparison with State-of-the-Art

We compare MATE (Motion-Aware Attention and Temporal-Conditioned Convolution Encoder) against recent state-of-the-art methods on UCF-Crime and XD-Violence. Experiments use two variants: MATE-BASE (40k parameters) and MATE-FAST (8.5k parameters), evaluated with I3D and X3D features. Metrics include ROC-AUC for UCF-Crime and Average Precision (AP) for XD-Violence, demonstrating performance and robustness across weakly supervised video anomaly and violence detection.

UCF-CRIME: FEATURE EXTRACTORS AND ROC-AUCs

Table 1. ROC-AUC (%) and feature extractors on UCF-Crime. † denotes results obtained by reproducing the original code.

Method	Feature	ROC-AUC (%)
Sultani et al. (Sultani et al., 2018)	C3D	75.41
Sultani et al. (Sultani et al., 2018)	I3D	77.92
MIST (Feng et al., 2022)	I3D	82.30
CLAWS (Zaheer et al., 2020)	C3D	83.03
RTFM (Jiang et al., 2021)	I3D	84.03
WSAL (Zaheer et al., 2020)	I3D	85.38
S3R (Öztürk & Can, 2021)	I3D	85.99
MGFN (Liu et al., 2022)	I3D	86.98
BN-WVAD (Zhou et al., 2023)	I3D	87.24
MATE-BASE(Ours)	I3D	89.05
MATE-FAST(Ours)	X3D	89.75
STEAD† (Gao & Liu, 2025)	X3D	91.34
MATE-BASE(Ours)	X3D	91.78

UCF-CRIME: ROC-AUC AND PARAMETERS

Table 2. ROC-AUC (%) and number of parameters on UCF-Crime. † denotes results obtained by reproducing the original code.

Method	ROC-AUC (%)	# Parameters
RTFM (Jiang et al., 2021)	84.03	24.72M
WSAL (Zaheer et al., 2020)	85.38	1.98M
S3R (Öztürk & Can, 2021)	85.99	0.33M
MGFN (Liu et al., 2022)	86.98	26.65M
BN-WVAD (Zhou et al., 2023)	87.24	5.79M
MATE-FAST(Ours)	89.75	8.5k
STEAD† (Gao & Liu, 2025)	91.34	1.63M
MATE-BASE(Ours)	91.78	40.1k

XD-VIOLENCE: AVERAGE PRECISION (AP)

Table 3. Comparison of Average Precision (AP %) on XD-Violence. † denotes results obtained by reproducing the original code.

Method	Feature	AP (%)
Sultani et al. (Sultani et al., 2018)	I3D	73.20
HL-Net (Zhou et al., 2020)	I3D	73.67
HL-Net (Zhou et al., 2020)	I3D+VGGish	78.64
RTFM (Jiang et al., 2021)	I3D	77.81
MSL (Zhang et al., 2022)	I3D	78.28
S3R (Öztürk & Can, 2021)	I3D	80.26
CU-Net (Zhou & et al., 2023a)	I3D	78.74
CU-Net (Zhou & et al., 2023a)	I3D+VGGish	81.43
UR-DMU (Zhou & et al., 2023b)	I3D	81.66
UR-DMU (Zhou & et al., 2023b)	I3D+VGGish	81.77
MACIL-SD (Liu, 2022)	I3D+VGGish	83.40
SAS (Chen, 2023)	I3D	83.59
BN-WVAD† (Zhou et al., 2023)	I3D	84.20
BN-WVAD (Zhou et al., 2023)	I3D	84.93
BN-WVAD (Zhou et al., 2023)	I3D+VGGish	85.26
MATE-BASE (Ours)	I3D	86.93

4.4.1. CROSS-VALIDATION RESULTS

For UCF-Crime, we perform stratified 5-fold cross-validation to assess stability under different data partitions using X3D features.

Table 4. 5-Fold Cross-Validation ROC-AUC (%) on UCF-Crime with X3D features.

Fold	Validation ROC-AUC (%)
1	89.98
2	92.25
3	91.73
4	93.27
5	92.40
Mean ± Std	91.926 ± 1.2

The 95% confidence interval (CI) of the mean val ROC-AUC is computed as $\bar{x} \pm t_{0.975, n-1} \cdot \frac{s}{\sqrt{n}}$, where n is the number of folds, s is the sample standard deviation, and $t_{0.975, n-1}$ is the t-statistic for 95% confidence. For our 5-fold cross-validation, the CI is approximately (90.44, 93.42)%.

The relatively narrow confidence interval demonstrates that MATE consistently achieves high performance across different data partitions, highlighting the stability and generalization capability of our model. This indicates that the model’s temporal decision-making is robust and not dependent on specific folds or outlier videos.

4.5. Ablation Studies

We conduct ablations to isolate the contributions of architectural components and hyperparameters while keeping the training protocol fixed. Metrics are reported as ROC-AUC on UCF-Crime.

EFFECT OF TEMPORAL QUERIES AND ATTENTION HEADS

We vary the number of learnable temporal queries and attention heads within the motion-aware pooling modules. Increasing the number of heads improves representation granularity, while a moderate number of queries avoids over-parameterization:

Table 5. Effect of number of queries and attention heads on ROC-AUC

Queries	Heads	ROC-AUC
8	8	91.77
8	4	89.52
6	8	89.43

COMPONENT-WISE ABLATION

Table 6. Component-wise ablation showing the impact of individual components on ROC-AUC and AP (%).

Removed Component	#Params	ROC-AUC (%)	AP (%)
None (Full Model)	40,130	91.77	92.30
Motion Map	40,130	90.00	89.17
Attention Block	35,970	90.70	89.59
TCC (w/o MS)	38,082	88.32	88.52
TCC (w/o MG)	40,097	89.59	89.30
TCC	26,689	90.27	90.58
TCC (w/o MS+MG)	38,049	89.00	89.00
TempPool*	39,874	87.94	89.14
Triplet Loss	40,130	89.18	88.23

MS denotes multiscale, MG denotes motion gating within the TCC module. *TempPool replaces the proposed motion-aware temporal pooling with standard max pooling.

Motion-Aware Interpretability and Analysis

MOTION MAP SENSITIVITY

We visualize the learned motion maps to understand the model’s attention to dynamic regions. As shown in Figure 2, the motion map for normal videos highlights only background or benign motion, while the map for anomalous videos strongly activates regions corresponding to abnormal events. The difference map (Anomaly—Normal) clearly emphasizes regions with increased motion, demonstrating that the motion-aware module effectively directs the network to focus on relevant spatiotemporal areas.

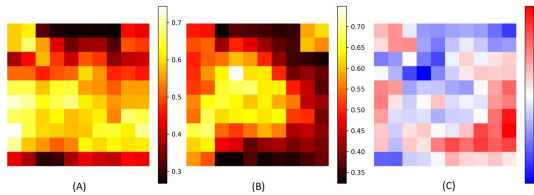


Figure 2. Motion map sensitivity: left-to-right visualization of (A) Normal, (B) Anomalous, and (C) Difference motion maps.

SNR Experiments

Robustness to noise is critical in real-world scenarios like autonomous driving and surveillance. We evaluate our model under realistic noise by adding Gaussian noise at different SNR levels Clean, 20, 10, 5, 0 dB.

NOISE SIMULATION

Let the clean video tensor be $\mathbf{X} \in \mathbb{R}^{B \times C \times T \times H \times W}$ and Gaussian noise

$$\mathbf{N} \sim \mathcal{N}(0, \sigma^2), \quad \tilde{\mathbf{X}} = \mathbf{X} + \mathbf{N}, \quad \sigma^2 = \frac{\text{signal_power}}{10^{\text{SNR}_{\text{dB}}/10}}.$$

Signal power is estimated as

$$\text{signal_power} = \frac{1}{BCTHW} \sum_{b,c,t,h,w} X_{b,c,t,h,w}^2.$$

NOISE MITIGATION VIA PREPROCESSING

Each frame is smoothed with a $K \times K$ averaging kernel, where $K = 3$ for $\text{SNR} > 5$ and $K = 5$ for $\text{SNR} \leq 5$:

$$\hat{X}_{b,c,t,h,w} = \frac{1}{K^2} \sum_{i=-\lfloor K/2 \rfloor}^{\lfloor K/2 \rfloor} \sum_{j=-\lfloor K/2 \rfloor}^{\lfloor K/2 \rfloor} \tilde{X}_{b,c,t,h+i,w+j}.$$

QUANTITATIVE RESULTS

Table 7. ROC-AUC performance under varying SNR levels.

SNR (dB)	MR	MPP	NMR	NMPP
Clean	91.78	91.78	91.27	91.27
20	91.73	91.09	90.88	90.80
10	86.32	90.08	85.42	91.08
5	81.97	90.00	81.26	89.80
0	76.85	86.55	75.94	86.42

MR: Motion Raw, MPP: Motion Preprocessed, NMR: No-Motion Raw, NMPP: No-Motion Preprocessed.

ROC BEHAVIOR WITH AND WITHOUT MOTION

Motion-aware pooling improves video-level anomaly detection, particularly in the low-FPR regime. The ROC curve in Figure 3 shows an AUC of 0.9178 with motion versus 0.900 without motion, demonstrating that motion cues enhance discriminative performance under weak supervision.

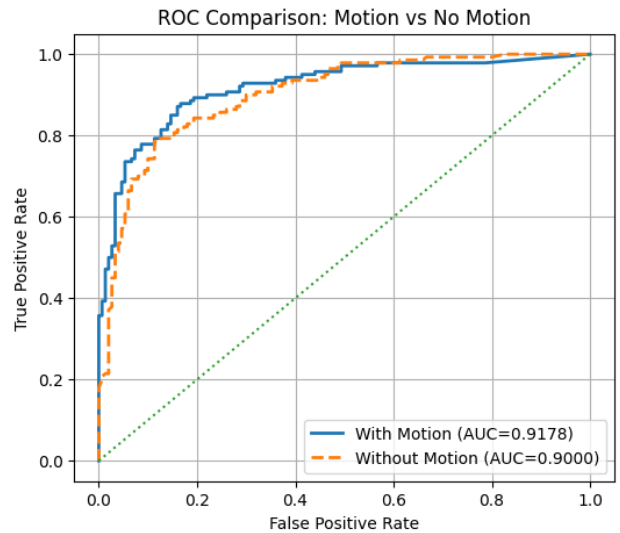


Figure 3. ROC comparison of models with and without motion-aware pooling.

EMBEDDING GEOMETRY (UMAP)

UMAP projections of the learned video features (Figure 4) show two main clusters: anomalous videos (red) mostly in the left-bottom, and normal videos (blue) in the right-top region. Motion-aware pooling improves cluster separation, reducing overlap and forming more discriminative manifolds, indicating that motion cues enhance embedding geometry under weak supervision.

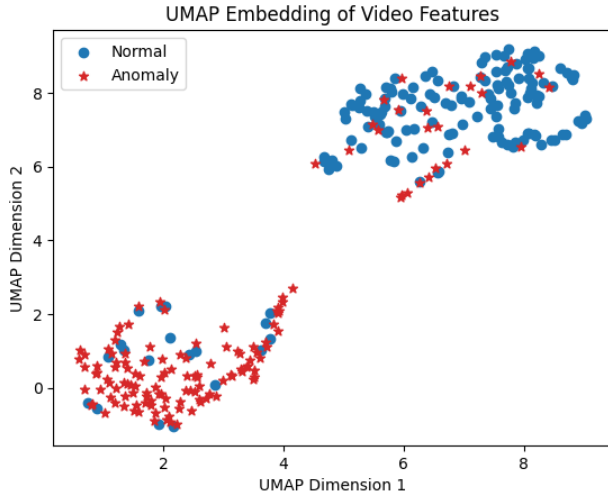


Figure 4. UMAP embedding of video features.

4.6. Discussion

We analyze the experimental results in Tables 1–6, along with noise robustness in Table 7, to evaluate the effectiveness, efficiency, and robustness of the proposed MATE architecture. On UCF-Crime (Table 1), MATE-BASE outperforms BN-WVAD by +1.81 ROC-AUC points using I3D features, and achieves 91.78% with X3D features, surpassing MATE-FAST and STEAD, confirming that *motion-aware temporal encoding provides complementary information to pre-extracted features*. Table 2 highlights the extreme parameter efficiency of MATE: while several baselines exceed 20M parameters, MATE-BASE uses only 40.1K parameters yet achieves the highest ROC-AUC, demonstrating that *lightweight attention and explicit motion modeling are more effective than increasing model size*. On XD-Violence (Table 3), MATE-BASE achieves 86.93% AP with visual features only, surpassing some audio-visual methods, indicating *motion-aware reasoning generalizes across datasets*.

Table 4 reports 5-fold cross-validation with X3D features, showing low variance (± 1.2 ROC-AUC) and confirming stable performance due to *motion-normalized attention weights and query-based temporal pooling*. Noise robustness (Table 7) shows that *motion-aware preprocessing (Motion PREP)* consistently mitigates degradation under varying

SNRs; for example, at 0 dB, Motion PREP retains 0.8655 ROC-AUC vs. 0.7685 for Motion RAW, while non-motion PREP also benefits, indicating that *temporal or feature normalization stabilizes predictions under noise*. Table 5 demonstrates that 8 attention heads and 8 temporal queries yield peak performance, confirming the importance of *attention diversity and query capacity*.

Ablation studies (Table 6) verify the contribution of each component: removing the *motion map*, disabling *multiscale temporal conditioning* or *motion gating*, removing the *Attention Block*, replacing motion-aware pooling with standard max pooling, or removing the triplet loss all degrade ROC-AUC and AP, highlighting that *each module causally contributes to performance*. While MATE achieves strong results with minimal parameters, it relies on pre-extracted features and does not perform end-to-end learning, suggesting future work could integrate motion-aware temporal encoding into backbone networks or explore multi-modal extensions for further generalization.

Limitations. While MATE achieves strong performance with minimal parameters, it relies on pre-extracted features and does not perform end-to-end feature learning. Future work could integrate motion-aware temporal encoding into backbone networks or explore multi-modal extensions to further enhance generalization.

5. Conclusion

We presented MATE, a lightweight, motion-aware temporal encoding architecture for weakly supervised video anomaly detection. Across benchmarks including UCF-Crime and XD-Violence, MATE outperforms prior methods while using far fewer parameters. Ablation studies confirm that motion maps, temporal conditioning, attention blocks, and motion-aware pooling each play a critical role in performance. Cross-validation demonstrates stable, reproducible results, and attention diversity and query capacity are essential for capturing heterogeneous anomalies.

MATE is also robust under noisy conditions, maintaining high ROC-AUC and AP even at low SNR levels using pre-processing, highlighting practical applicability in real-world surveillance. Overall, incorporating explicit motion-aware inductive biases with efficient attention enables strong, interpretable anomaly detection without large computational overhead. Future work may explore end-to-end backbone integration or multi-modal extensions to further improve generalization and real-world deployment.

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A. Appendix

This appendix provides a unified theoretical explanation of the proposed architecture, clarifying the design motivations behind each component and their interaction. The goal of this appendix is to justify why the proposed design constitutes a principled and effective spatiotemporal modeling approach.

A.1. Notation and Problem Setup

Let

$$x_t \in \mathbb{R}^{H \times W \times C}, \quad t \in \{1, \dots, T\}$$

denote the learned feature representation of a video frame at time t . A video sequence is modeled as a discrete-time dynamical system:

$$x_{t+1} = x_t + \Delta_t,$$

where Δ_t captures spatiotemporal variation.

The model implements a parameterized operator

$$\mathcal{F}_\theta : \{x_t\}_{t=1}^T \rightarrow y,$$

designed to emphasize motion-salient regions while maintaining computational efficiency and stability.

A.2. Motion Map as a Normalized Temporal Derivative

Definition A.1 (Motion Map). The motion map is computed as:

$$m_t = \frac{\left\| x_t - \frac{1}{K+1} \sum_{i=0}^K x_{t-i} \right\|_2}{\max_{t'} \left\| x_{t'} - \frac{1}{K+1} \sum_{i=0}^K x_{t'-i} \right\|_2 + \varepsilon}.$$

Proposition A.1 (Finite-Difference Approximation). Assume that $x(t)$ is temporally differentiable. Then:

$$x_t - \frac{1}{K+1} \sum_{i=0}^K x_{t-i} = \mathcal{O}\left(\frac{\partial x}{\partial t}\right).$$

Proof Sketch. Using Taylor expansion around t :

$$x_{t-i} = x_t - i\Delta t \frac{\partial x}{\partial t} + \mathcal{O}(\Delta t^2).$$

Averaging over i cancels higher-order terms, yielding a first-order temporal derivative estimate. $\square$

Remark A.1 (Boundedness). The normalization ensures $m_t \in [0, 1]$, bounding all motion-dependent modulations used in subsequent convolution, attention, and pooling operations.

A.3. Temporal Conditioned Convolution as a Dynamic Operator

Temporal Conditioned Convolution (TCC) computes:

$$y_t = \text{Conv}(x_t) \odot \sigma(Wg(x_{1:T})),$$

where $g(\cdot)$ aggregates temporal statistics.

Proposition A.2 (Approximation of Dynamic Convolution). TCC approximates a time-varying convolution operator:

$$\text{Conv}_{\theta(t)}(x_t) \approx \text{Conv}_{\theta_0}(x_t) + \frac{\partial \theta}{\partial t} \Delta t.$$

Proof Sketch. Conditioning convolution parameters on temporal context introduces a first-order perturbation to static kernels, corresponding to a truncated Taylor expansion of a fully dynamic convolution. $\square$

Motion-Gated Residual Update. The residual update is given by:

$$x_t \leftarrow x_t + \alpha g_t y_t, \quad g_t = \sigma(W_g h_t + m_t),$$

where motion magnitude modulates update strength, suppressing updates in static regions.

A.4. Efficient Dynamic Attention via Kernelized Linear Attention

The attention module employs the positive feature map:

$$\phi(x) = \text{ELU}(x) + 1.$$

Proposition A.3 (Kernelized Linear Attention). Given queries Q , keys K , and values V , attention is computed as:

$$\text{Attn}(Q, K, V) = \phi(Q) (\phi(K)^\top V) (\phi(Q) \phi(K)^\top \mathbf{1})^{-1}.$$

This corresponds to attention under the positive semi-definite kernel:

$$k(q, k) = \phi(q)^\top \phi(k),$$

as introduced in kernelized linear attention (Katharopoulos et al., 2020a).

Motion-Weighted Queries. Queries are modulated as:

$$Q' = Q \odot (1 + \tanh(m_t)).$$

Proposition A.4 (Motion Saliency Bias). Under positive kernel attention, motion-weighted queries monotonically increase attention mass assigned to motion-salient regions:

$$\frac{\partial \text{Attn}}{\partial m_t} > 0,$$

which follows from the monotonicity of kernelized attention (Katharopoulos et al., 2020a).

A.5. Motion-Aware Temporal Pooling and Permutation Invariance

Spatial features are aggregated as:

$$\hat{x}_t = \sum_i \alpha_{t,i} x_{t,i}, \quad \alpha_{t,i} \propto m_{t,i}.$$

Temporal aggregation is performed using learnable queries.

Proposition A.5 (Permutation-Invariant Aggregation). The pooling operation satisfies the DeepSets decomposition:

$$f(\{x_t\}) = \rho \left(\sum_t \phi(x_t, m_t) \right),$$

and is permutation invariant with respect to temporal ordering, consistent with the DeepSets characterization (Zaheer et al., 2017).

A.6. Model as a Stable Dynamical System

The full block update can be written as:

$$x_{t+1} = x_t + \mathcal{F}_\theta(x_t, m_t),$$

where $\mathcal{F}_\theta$ includes temporal conditioned convolution, kernelized attention, and feed-forward updates.

Proposition A.6 (Lipschitz Stability). Assuming bounded parameters and normalized motion maps, there exists $L < \infty$ such that:

$$\|\mathcal{F}_\theta(x) - \mathcal{F}_\theta(y)\| \leq L\|x - y\|.$$

Proof Sketch. Each component is Lipschitz-continuous under bounded weights, and the composition of Lipschitz functions remains Lipschitz-continuous (Anil et al., 2019). $\square$

A.7. Summary

This analysis shows that the proposed architecture: (i) approximates first-order temporal dynamics in learned feature space, (ii) uses motion as a bounded global control signal, (iii) employs kernelized linear attention for efficiency, (iv) ensures permutation-invariant temporal aggregation, and (v) forms a stable residual spatiotemporal dynamical system.